

SMART INDIA HACKATHON 2026

Idea / Proposal Submission Report

KrishiSetu

“A Digital Bridge Connecting Farmers Directly to Consumers”

Theme: Agriculture, FoodTech & Rural Development

Problem Statement: Multiple Intermediaries Reduce Farmers' Earnings and Increase Consumer Prices

[Team Name – PARALLAX]

[Team ID – 152023]



1. Title & Team Details

1.1 Project Title

KrishiSetu – An AI-Enabled Direct Farmer-to-Consumer Agri-Marketplace Reducing Intermediary Dependence.

1.2 Team Details

Table 1

S.No.	Name	Role	Institute	Course
1	Priyanshu Singh	Team Leader	Institute Of Technology And Management, GIDA, Gorakhpur	B.Tech
2	Ayush Mishra	Member 1	Institute Of Technology And Management, GIDA, Gorakhpur	B.Tech
3	Prajawal Mishra	Member 2	Institute Of Technology And Management, GIDA, Gorakhpur	B.Tech
4	Krishna	Member 3	Institute Of Technology And Management, GIDA, Gorakhpur	B.Tech
5	Alankriti Choudhary	Member 4	Institute Of Technology And Management, GIDA, Gorakhpur	BCA
6	Deepali Mishra	Member 5	Institute Of Technology And Management, GIDA, Gorakhpur	BCA

2. Problem Statement

India's agricultural supply chain is dominated by a long chain of intermediaries — commission agents (arhatiyas), wholesalers, sub-wholesalers, and multiple layers of retailers — between the farmer and the final consumer. At each stage, margins are added without proportionate value addition, resulting in a situation where farmers receive only a small fraction of the price paid by the end consumer, while consumers pay significantly inflated prices for the same produce. This price gap, often referred to as the 'farm-to-fork' price spread, undermines farmer income, discourages investment in better farming practices, and contributes to food price volatility for urban and rural consumers alike. The problem statement calls for a technology-driven solution that shortens or restructures this supply chain, ensures price transparency, and creates a fair, efficient, and trustworthy marketplace connecting farmers directly with consumers, retailers, and bulk buyers.

3. Problem Background & Current Challenges

3.1 Background

Traditional agricultural marketing in India routes the majority of produce through regulated mandis (APMC yards) and unregulated local markets, where 3 to 6 intermediary layers are typical before produce reaches the consumer. Studies by government and industry bodies have repeatedly shown that farmers realise only 25–40% of the final consumer price for perishable commodities such as fruits and vegetables, with the remaining share absorbed by commission agents, transporters, wholesalers, and retailers. Despite reforms such as the model APMC Act amendments, e-NAM, and the Farmers' Produce Trade and Commerce Act, adoption of digital and direct-marketing alternatives remains low, especially among small and marginal farmers.

3.2 Current Challenges

- Multiple layers of commission agents and wholesalers each add a margin, inflating final prices without adding proportional value.
- Lack of price transparency — farmers often do not know the prevailing market or retail price of their produce and are forced to accept whatever rate the local trader offers.
- Distress selling due to lack of storage and cold-chain infrastructure, forcing farmers to sell immediately after harvest at low prices.
- Delayed and uncertain payments from traditional mandis and middlemen, leading to cash-flow stress for farmers.
- Fragmented and inefficient logistics resulting in high transportation costs and post-harvest losses of 15–30% for perishable produce.
- Low digital literacy and limited smartphone/internet access among small and marginal farmers, limiting adoption of existing agri-tech platforms.
- Absence of trust and grading/quality-standardisation mechanisms between unfamiliar farmers and consumers/buyers.
- Consumers pay high retail prices despite farmers receiving low farm-gate prices, indicating inefficiency rather than fair value distribution.

4. Proposed Solution – KrishiSetu

KrishiSetu is proposed as a mobile and web-based digital marketplace platform that directly connects farmers with consumers, retailers, restaurants, and bulk institutional buyers, eliminating unnecessary layers of intermediaries while retaining essential support services such as logistics, quality assurance, and payment security through the platform itself. Rather than removing all intermediation (which is often practically necessary for aggregation, transport, and quality control), KrishiSetu replaces the fragmented, opaque, multi-layer intermediary chain with a single, transparent, technology-managed intermediation layer — ensuring farmers get a fair, visible price and consumers get fresh produce at a reasonable cost.

The platform uses AI-driven price discovery and demand forecasting, an integrated logistics and route-optimisation engine, and a conditional/escrow-based payment system to build trust between previously unfamiliar parties, while supporting farmers with multilingual, low-bandwidth, and voice-assisted interfaces to overcome digital literacy barriers.

5. Objectives

- To reduce the number of intermediary layers between farmer and consumer, thereby increasing the farmer's share of the final consumer price.
- To provide real-time, transparent price discovery based on mandi rates, demand-supply trends, and AI-based forecasting.
- To build an efficient, optimised logistics network for last-mile and bulk transportation of produce, minimising spoilage and cost.
- To ensure secure and timely payments to farmers through a conditional/escrow-based digital payment mechanism.
- To create an accessible, farmer-friendly interface supporting regional languages and voice input for low digital-literacy users.
- To enable data-driven decision-making for farmers regarding crop selection, pricing, and selling time through analytics and insights.
- To create a scalable, replicable model that can be extended across states, crop categories, and buyer segments (B2C, B2B, institutional).

6. How KrishiSetu Works / System Workflow

The KrishiSetu workflow is designed to take produce from the farm to the buyer through a minimal number of clearly defined, technology-mediated steps:

Step 1: Farmer Onboarding & Listing

Farmers register on the app/web portal (or through a local KrishiSetu kiosk/agent for assisted onboarding), verify their identity and land details, and list available produce with quantity, expected harvest date, and self-reported quality grade, optionally aided by photo-based AI quality assessment.

Step 2: AI-Based Price Recommendation

The system fetches live mandi prices (e.g., via e-NAM/Agmarknet APIs), analyses historical trends and current demand signals, and recommends a fair price range to the farmer, which the farmer can accept or adjust within platform guardrails.

Step 3: Matching with Buyers

Consumers, retailers, restaurants, and bulk/institutional buyers browse or are auto-matched with nearby available produce based on location, quantity needs, and price, through a demand-aggregation engine that batches small consumer orders into efficient bulk pickups.

Step 4: Order Confirmation & Conditional Payment

On order confirmation, the buyer's payment is collected and held in a conditional escrow. Funds are released to the farmer automatically once pickup/delivery and quality confirmation are completed, protecting both parties from default or dispute.

Step 5: Logistics & Delivery

The route-optimisation engine assigns the nearest available KrishiSetu logistics partner (aggregator vehicle, local delivery partner, or partnered transport) to collect produce from the farm/collection point and deliver it to the buyer or nearest fulfilment/retail point within the shortest optimised route and time.

Step 6: Feedback, Rating & Analytics

After delivery, both farmer and buyer rate the transaction. This feedback, along with sales data, feeds back into the AI models to refine price recommendations, quality scoring, and demand forecasts for future cycles.

7. Technical Approach & Tech Stack

7.1 Overall Architecture Approach

KrishiSetu follows a modular, microservices-based architecture with separate services for user management, listings, price-recommendation (AI/ML), order and payment management, logistics/routing, and analytics — communicating through REST/GraphQL APIs and a message queue for asynchronous events such as order status updates and payment confirmations.

7.2 Proposed Tech Stack

Table 2

Layer	Technology
Architecture	Modular service-based design connecting frontend clients to backend APIs, AI/ML services, databases, logistics/routing, payment and notification modules.
Frontend	HTML, CSS, JavaScript, Bootstrap
Backend	Python Flask + REST API
Database	PostgreSQL + SQLAlchemy
AI/ML	Pandas, NumPy, Scikit-learn

Layer	Technology
Maps & Routing	Leaflet, OpenStreetMap, OSRM
Optimisation	Google OR-Tools
Payments	Razorpay

8. AI / ML Components

Demand Forecasting

Models analyse historical orders, seasonal patterns and regional demand signals to estimate upcoming crop demand.

Price Recommendation

Market prices, historical trends and demand-supply signals can be used to recommend a competitive price range while allowing farmer control.

Quality Assistance

Image-based analysis can assist in identifying visible quality characteristics such as size, ripeness and defects.

Personalisation

Buyer recommendations can use past orders and preferences; farmer insights can use regional trends and sales history.

9. Logistics & Route Optimization

KrishiSetu's logistics module aggregates nearby orders into optimised pickup and delivery routes rather than dispatching a separate trip per transaction, significantly reducing per-unit transportation cost. The route-optimisation engine uses a vehicle-routing-problem (VRP) formulation — solved using tools such as Google OR-Tools — that considers pickup locations (farms/collection centres), delivery locations (consumers, retail points, hubs), vehicle capacity, produce perishability windows, and traffic/distance data to generate the most time- and cost-efficient route for each logistics partner.

- Collection-centre model: small quantities from nearby farmers are aggregated at local collection points before onward bulk transport.
- Partnered fleet and gig-delivery network for last-mile delivery in urban areas.
- Real-time tracking for both farmers and buyers to monitor pickup and delivery status.
- Cold-chain partner integration (where available) for highly perishable produce to minimise post-harvest losses.

10. Database & System Architecture

The system architecture follows a layered, microservices design: a client layer (mobile/web apps for farmers, buyers, and admin), an API gateway layer for authentication and routing, a services layer

(user, listing, pricing/AI, order, payment, logistics, notification microservices), and a data layer combining relational storage (PostgreSQL) for transactional data such as users, orders, and payments with a document store (MongoDB) for flexible produce-listing catalogues, plus Redis for caching frequently accessed data such as live prices.

Core Database Entities (Illustrative)

- Users (Farmers, Buyers, Logistics Partners, Admins) — profile, KYC, location, trust score.
- Produce Listings — crop type, quantity, quality grade, price, harvest date, farmer reference.
- Orders — buyer, listing reference(s), quantity, price, status, timestamps.
- Payments/Escrow — order reference, amount, status (held/released/refunded), transaction ID.
- Logistics Jobs — pickup/drop points, assigned partner, route, status.
- Price History & Market Data — crop, mandi, region, date, price (for AI training).
- Ratings & Feedback — transaction reference, rating, comments.

11. User Roles & Modules

Table 3

Role	Key Modules / Capabilities
Farmer	Registration/KYC, produce listing, AI price suggestion, order management, earnings dashboard, payment status, crop advisory & analytics.
Consumer (B2C)	Browse/search produce, place orders, track delivery, rate farmer/produce, view price transparency breakdown.
Retailer / Restaurant (B2B)	Bulk ordering, recurring/subscription orders, invoicing, quality specification requests.
Logistics Partner	Route assignment, pickup/delivery status updates, real-time tracking, proof-of-delivery capture.
Admin	User/KYC verification, dispute resolution, platform analytics, commission & payout management, content moderation.

12. Risks & Mitigation

Table 4

Risk	Mitigation Strategy
Low digital literacy / smartphone access among farmers	Voice-assisted, multilingual UI; assisted onboarding via local kiosks/agents and FPO partnerships.
Resistance from existing middlemen/mandi ecosystem	Position select intermediaries as logistics/collection-centre partners rather than fully excluding them; phased rollout.

Risk	Mitigation Strategy
Logistics failure / produce spoilage in transit	Route optimisation, collection-centre aggregation, cold-chain partnerships for perishables, insurance coverage.
Trust deficit between unfamiliar farmers and buyers	Escrow-based conditional payments, quality grading via AI, rating/trust-score system.
Price volatility affecting AI recommendations	Continuous model retraining with live mandi data; human-override option for farmers.
Low initial buyer-side adoption (chicken-and-egg problem)	Pilot launch with FPOs, RWAs, and institutional buyers (restaurants/canteens) to seed early demand and supply.
Data privacy & security concerns	Encrypted storage, role-based access control, compliance with applicable data-protection regulations.

13. Impact & Benefits

13.1 For Farmers

- Higher realised price per unit of produce due to fewer margin-taking layers.
- Transparent, predictable, and timely payments through escrow-based settlement.
- Reduced distress selling through better price visibility and demand matching.
- Access to data-driven insights for better crop and pricing decisions.

13.2 For Consumers

- Lower prices for fresh produce due to reduced cumulative middleman margins.
- Greater freshness and traceability of purchased produce.
- Transparency into how much of the price reaches the farmer.

13.3 For the Broader Ecosystem

- Reduced post-harvest wastage through optimised logistics and demand matching.
- Support for government goals of doubling farmer income and promoting digital agriculture.
- Formal digital footprint for farmers, enabling future access to credit and insurance products.

14. Innovation / Key Differentiators

- Combines AI-based dynamic pricing with real mandi-data grounding, rather than static or purely negotiated pricing.
- Escrow-based conditional settlement specifically designed for trust-building between first-time, unfamiliar farmer-buyer pairs.

- Demand-aggregation logistics model that makes last-mile delivery of small farm orders economically viable, unlike typical single-order delivery apps.
- Inclusive design (voice, multilingual, kiosk-assisted onboarding) targeted specifically at low digital-literacy farmers, not just tech-savvy early adopters.
- Dual-sided value proposition — simultaneously increasing farmer income and reducing consumer prices — rather than optimising for only one side of the market.

15. Survey & Primary Research

To validate the problem and proposed solution, the team conducted primary research through a structured Google Form survey circulated among farmers, consumers, and local vendors, supplemented by direct interviews with farmers in [region/village name] to understand their current selling practices, pain points, and willingness to adopt a digital direct-marketing platform

16. Google Form Screenshot [\[LINK\]](#)

The following screenshot shows the structure of the survey form used to gather primary data from farmers and consumers regarding current selling/buying practices and pricing perceptions.

KrishiSetu – Agricultural Supply Chain Survey

We are conducting a short survey to understand the challenges faced by farmers, agricultural intermediaries, and consumers/buyers in the current agricultural supply chain. Your responses will help us understand real-world problems related to pricing, market access, transportation, wastage, and purchasing.

हम वर्तमान कृषि आपूर्ति श्रृंखला में किसानों, कृषि मध्यस्थों और उपभोक्ताओं/खरीदारों के सामने आने वाली चुनौतियों को समझने के लिए एक छोटा सर्वेक्षण कर रहे हैं। आपकी प्रतिक्रियाएँ हमें मूल्य निर्धारण, बाजार तक पहुँच, परिवहन, बर्बादी और खरीदारी से जुड़ी वास्तविक समस्याओं को समझने में मदद करेंगी।

psingh302006@gmail.com [Switch account](#)

Not shared

* Indicates required question

KrishiSetu – Connecting Farmers to Better Markets



Which category best describes you? *

आप निम्नलिखित में से किस श्रेणी से संबंधित हैं?

☐ Farmer / FPO Member – किसान / FPO सदस्य

☐ Trader / Middleman – व्यापारी / विचरित्र

☐ Consumer – उपभोक्ता

☐ Bulk Buyer / Retailer – थोक खरीदार / खुदरा विक्रेता

☐ Other: _____

Where do you usually buy/sell agricultural produce? *

आप आमतौर पर कृषि उपज कहाँ खरीदते/बेचते हैं?

☐ Local Market / Mandi – स्थानीय बाजार / मंडी

☐ Directly to Buyers – सीधे खरीदारों को

☐ Through Traders / Middlemen – व्यापारियों / विचरित्रों के माध्यम से

☐ Wholesale Market – थोक बाजार

☐ Retail Shop / Local Vendor – खुदरा दुकान / स्थानीय विक्रेता

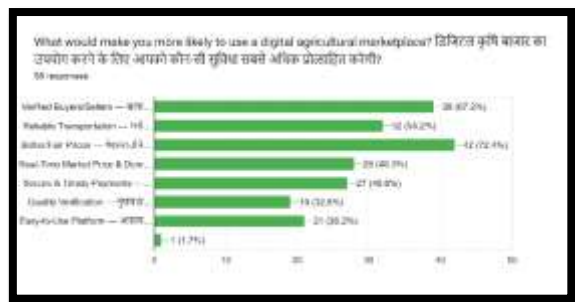
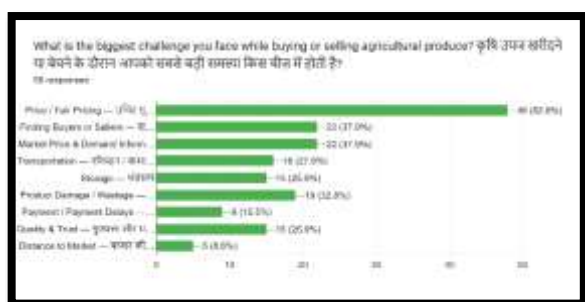
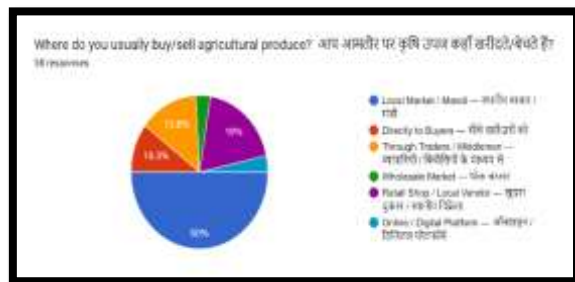
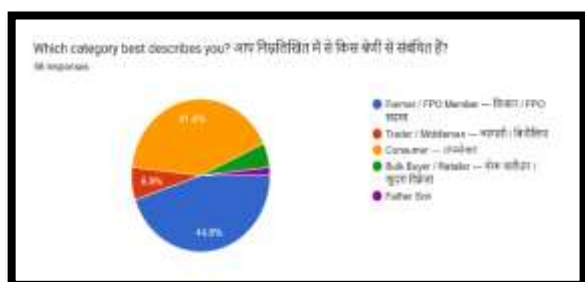
☐ Online / Digital Platform – ऑनलाइन / डिजिटल प्लेटफॉर्म

☐ Other: _____

What is the biggest challenge you face while buying or selling agricultural produce? *

17. Actual Response Charts/Results [\[LINK\]](#)

Key findings from the survey responses are summarised below, illustrating the extent of the intermediary problem and the level of interest in a direct digital marketplace among respondents.



18. Farmer Interview

The team conducted on-ground/video interviews with farmer to gather qualitative insights on current market practices, pricing challenges, and expectations from a digital solution.

[SURVEY VIDEO](#)

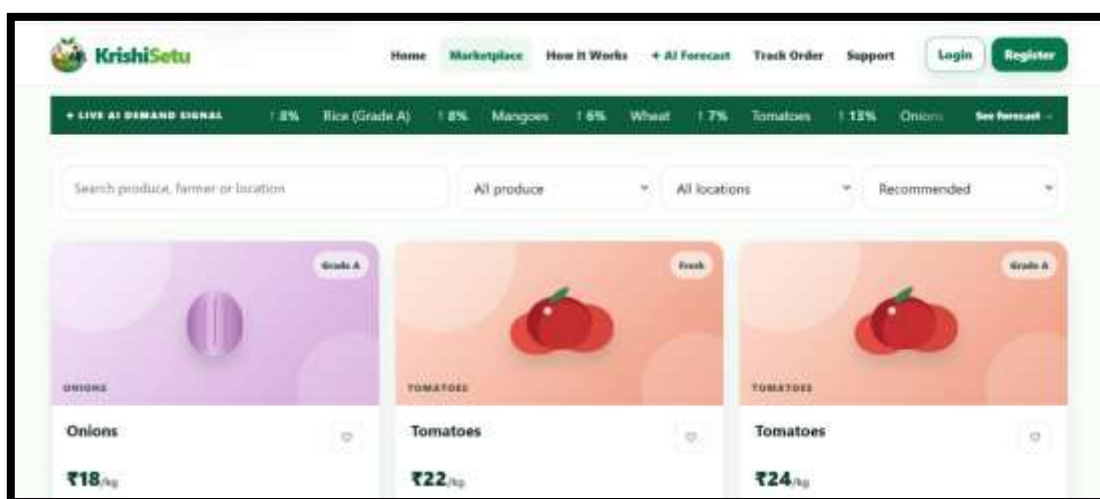
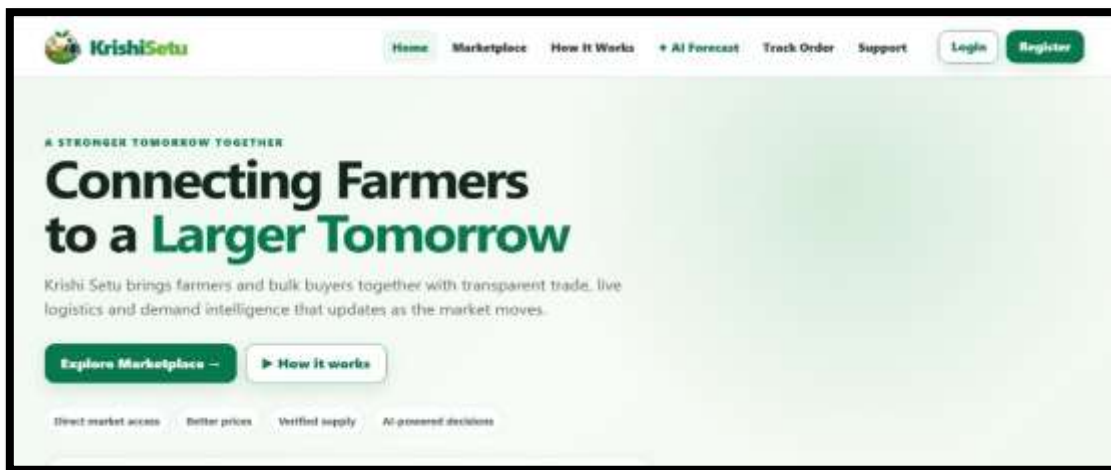
19. Competitor / Existing Platform Comparison

Parameter	KrishiSetu	e-NAM	DeHaat	Local Mandi/Middlemen
Direct farmer-to-consumer sale	Yes	Limited (B2Bfocus)	Limited	No
AI-based price recommendation	Yes	No	Partial	No
Escrow/conditional payment	Yes	No	Partial	No
Integrated logistics & route optimisation	Yes	No	Partial	No
Voice/multilingual low-literacy support	Yes	Limited	Partial	N/A
Consumer-facing app (B2C)	Yes	No	No	No

Note: The above comparison is illustrative and should be validated/updated with the latest feature sets of each platform before final submission.

20. Prototype Screenshots

The following screenshots showcase the current prototype/UI mockups of the KrishiSetu application across key user journeys.



21. Prototype Demo Video Link

Video Link:

[\[LINK\]](#)

22. Future Scope

- Expansion to additional states, crops (grains, dairy, poultry), and buyer categories (large institutional/export buyers).
- Integration with government schemes (PM-KISAN, crop insurance, KCC) for a unified farmer digital identity.
- Blockchain-based traceability for premium/organic produce certification.
- Partnerships with cold-chain and warehousing providers to further reduce post-harvest losses.
- Credit-scoring and micro-lending features for farmers based on platform transaction history.
- Advanced satellite/IoT-based crop health monitoring integrated with price and yield forecasting.

24. Limitations

- Initial dependence on internet/smartphone connectivity in remote rural areas, despite low-bandwidth design.
- Logistics viability may be lower in sparsely populated or poorly connected regions during early rollout.
- AI price recommendations require sufficient historical/live mandi data, which may be limited for certain regional or minor crops.
- Behavioural change required from farmers accustomed to traditional, relationship-based selling through local traders.
- Cold-chain and last-mile infrastructure gaps may limit handling of highly perishable produce in the initial phase.

25. Conclusion

The presence of multiple, opaque intermediary layers in India's agricultural supply chain simultaneously depresses farmer incomes and inflates consumer prices, representing a significant inefficiency rather than a necessary market feature. KrishiSetu addresses this problem through a technology-enabled, transparent, single-layer intermediation model — combining AI-driven price discovery, optimised aggregated logistics, and trust-building escrow payments — to create a fairer, more efficient agricultural marketplace. By focusing on inclusive design for low digital-literacy farmers alongside a strong consumer/retailer value proposition, KrishiSetu is positioned as a scalable, sustainable solution aligned with national goals of improving farmer welfare and agricultural market efficiency.

26. References / Sources

- RBI (2024) – farmer share in consumer prices.
[\[LINK\]](#)
- MoFPI / NABCONS (2022) – post-harvest loss estimates.
[\[LINK\]](#)
- Agricultural Economics Research Review (2024) – digital agricultural platforms and market access.
[\[LINK\]](#)
- TRAI (2024) – rural internet subscriber data.
[\[LINK\]](#)
- Government of India – e-NAM and agricultural digital infrastructure.
[\[LINK\]](#)
- Agriculture Census – operational holdings and size-class data.
[\[LINK\]](#)
- AgriStack – Ministry of Agriculture & Farmers Welfare.
[\[LINK\]](#)
- Primary survey and interview data collected by the team.
